| Operator: | Matrix A | Matrix B | Result | |
|---|---|---|---|---|
| + | | | 0 | 0 |
| | | | 0 | 0 |
| | | | 0 | 0 |
| TRANSPOSE | | | 0 | 0 |
| | | | 0 | 0 |
| | | | 0 | 0 |
| MDETERM | | | #VALUE! | |
| MINVERSE | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| MMULT | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| + | 1 | 1 | 2 | 0 |
| | | | 0 | 0 |
| | | | 0 | 0 |
| TRANSPOSE | 1 | 1 | 1 | 0 |
| | | | 0 | 0 |
| | | | 0 | 0 |
| MDETERM | 1 | 1 | #VALUE! | |
| MINVERSE | 1 | 1 | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| MMULT | 1 | 1 | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| | | | #VALUE! | #VALUE! |
| + | 1 | 1 | 2 | 2 |
| | | | 2 | 2 |
| | | | 2 | 2 |
| TRANSPOSE | 1 | 1 | 1 | 1 |
| | | | 1 | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | 1 | 1 |
| MDETERM | 1 | | 1 | | 1 | |
| MINVERSE | 1 | | 1 | | 1 | 1 |
| | | | | | 1 | 1 |
| | | | | | 1 | 1 |
| MMULT | 1 | | 1 | | 1 | 1 |
| | | | | | 1 | 1 |
| | | | | | 1 | 1 |
| + | 1 | 2 | 1 | 2 | 2 | 4 |
| | | | | | 2 | 4 |
| | | | | | 2 | 4 |
| TRANSPOSE | 1 | 2 | 1 | 2 | 1 | 1 |
| | | | | | 2 | 2 |
| | | | | | #N/A | #N/A |
| MDETERM | 1 | 2 | 1 | 2 | #VALUE! | |
| MINVERSE | 1 | 2 | 1 | 2 | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| MMULT | 1 | 2 | 1 | 2 | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| + | 1 | | 1 | | 2 | 2 |
| | 2 | | 2 | | 4 | 4 |
| | | | | | #N/A | #N/A |
| TRANSPOSE | 1 | | 1 | | 1 | 2 |
| | 2 | | 2 | | 1 | 2 |
| | | | | | 1 | 2 |
| MDETERM | 1 | | 1 | | #VALUE! | |
| | 2 | | 2 | | | |
| MINVERSE | 1 | | 1 | | #VALUE! | #VALUE! |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 2 | | 2 | | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| MMULT | 1 | | 1 | | #VALUE! | #VALUE! |
| | 2 | | 2 | | #VALUE! | #VALUE! |
| | | | | | #VALUE! | #VALUE! |
| + | 1 | 2 | 1 | 2 | 2 | 4 |
| | 2 | 3 | 2 | 3 | 4 | 6 |
| | | | | | #N/A | #N/A |
| TRANSPOSE | 1 | 2 | 1 | 2 | 1 | 2 |
| | 2 | 3 | 2 | 3 | 2 | 3 |
| | | | | | #N/A | #N/A |
| MDETERM | 1 | 2 | 1 | 2 | -1 | |
| | 2 | 3 | 2 | 3 | | |
| MINVERSE | 1 | 2 | 1 | 2 | -3 | 2 |
| | 2 | 3 | 2 | 3 | 2 | -1 |
| | | | | | #N/A | #N/A |
| MMULT | 1 | 2 | 1 | 2 | 5 | 8 |
| | 2 | 3 | 2 | 3 | 8 | 13 |
| | | | | | #N/A | #N/A |
| MMULT | 1 | 2 | 1 | | 5 | 5 |
| | | | 2 | | 5 | 5 |
| | | | | | 5 | 5 |
| MMULT | 1 | | 1 | 2 | 1 | 2 |
| | 2 | | | | 2 | 4 |
| | | | | | #N/A | #N/A |
| + | 1 | 2 | 1 | 2 | 2 | 4 |
| | 3 | 4 | 3 | 4 | 6 | 8 |
| | | | | | #N/A | #N/A |
| TRANSPOSE | | | | | 1 | 3 |
| | | | | | 2 | 4 |
| | | | | | #N/A | #N/A |

<END>

| | | Expected | | Test Params: | Test Nc |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | Input: One Entry, NULL | MDETERM: Only c |
| 0 | 0 | 0 | 0 | Output: 3x3 | MMULT: |
| 0 | 0 | 0 | 0 | | |
| | | | | | |
| 0 | 0 | 0 | 0 | | |
| 0 | 0 | 0 | 0 | | |
| 0 | 0 | 0 | 0 | | |
| | | | | | |
| | #VALUE! | | | <-Singular Array Formula Out | |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| | | | |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | | |
|---|---|---|---|---|
| 0 | 2 | 0 | 0 | Input: 3x3, One Entry |
| 0 | 0 | 0 | 0 | Output: 3x3 |
| 0 | 0 | 0 | 0 | |
| | | | | |
| 0 | 1 | 0 | 0 | |
| 0 | 0 | 0 | 0 | |
| 0 | 0 | 0 | 0 | |
| | | | | |
| | #VALUE! | | | <-Singular Array Formula Out |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| | | | |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | | |
|---|---|---|---|---|
| 2 | 2 | 2 | 2 | Input: Scalar |
| 2 | 2 | 2 | 2 | Output: 3x3 |
| 2 | 2 | 2 | 2 | |
| | | | | |
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | |

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | |
| | 1 | | | <- Scalar Out |

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | |
| | | | | |
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | |
| | | | | |
| #N/A | 2 | 4 | #N/A | Input: 1x2 |
| #N/A | 2 | 4 | #N/A | Output: 3x3 |
| #N/A | 2 | 4 | #N/A | |
| | | | | |
| 1 | 1 | 1 | 1 | |
| 2 | 2 | 2 | 2 | |
| #N/A | #N/A | #N/A | #N/A | |
| | #VALUE! | | | <- Scalar Out |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| | | | |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | | |
|---|---|---|---|---|
| 2 | 2 | 2 | 2 | Input: 2x1 |
| 4 | 4 | 4 | 4 | Output: 3x3 |
| #N/A | #N/A | #N/A | #N/A | |
| | | | | |
| #N/A | 1 | 2 | #N/A | |
| #N/A | 1 | 2 | #N/A | |
| #N/A | 1 | 2 | #N/A | |
| | | | | |
| | #VALUE! | | | <- Scalar Out |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | |
|---|---|---|---|
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |
| #VALUE! | #VALUE! | #VALUE! | #VALUE! |

| | | | | |
|---|---|---|---|---|
| #N/A | 2 | 4 | #N/A | Input: 2x2 |
| #N/A | 4 | 6 | #N/A | Output: 3x3 |
| #N/A | #N/A | #N/A | #N/A | |

| | | | |
|---|---|---|---|
| #N/A | 1 | 2 | #N/A |
| #N/A | 2 | 3 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

| | |
|---|---|
| -1 | <- Scalar Out |

| | | | |
|---|---|---|---|
| #N/A | -3 | 2 | #N/A |
| #N/A | 2 | -1 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

| | | | |
|---|---|---|---|
| #N/A | 5 | 8 | #N/A |
| #N/A | 8 | 13 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

| | | | | |
|---|---|---|---|---|
| 5 | 5 | 5 | 5 | Input: 2x2 |
| 5 | 5 | 5 | 5 | Output: 3x3 |
| 5 | 5 | 5 | 5 | Compatible MMULT (NOT SQUARE) |

| | | | |
|---|---|---|---|
| #N/A | 1 | 2 | #N/A |
| #N/A | 2 | 4 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

| | | | |
|---|---|---|---|
| #N/A | 2 | 4 | #N/A |
| #N/A | 6 | 8 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

| | | | |
|---|---|---|---|
| #N/A | 1 | 3 | #N/A |
| #N/A | 2 | 4 | #N/A |
| #N/A | #N/A | #N/A | #N/A |

ɔtes:
ɔutputs a scalar
: Must